

Lilycrest Dormitory Management System (Lilycrest DMS)

Software Requirements Specification, Agile User Stories Catalog, and Master
Quality Assurance Test Plan



PROJECT LEVEL

3rd Year Capstone System

DOCUMENT VERSION

Version 1.0 (Final Production-Ready Release)

OPERATIONAL SCOPE

Full Tenant Lifecycle (Reservation to Offboarding)

TARGET ENVIRONMENT

Multi-Branch (Gil Puyat & Guadalupe)

TESTING FRAMEWORK

Jest 30.x / Playwright / Memory Server

RELEASE DATE

September 2026

1. Executive Summary & Access Hierarchy

The **Lilycrest Dormitory Management System (Lilycrest DMS)** is a centralized web platform engineered to automate multi-branch operations across student and professional housing facilities in Metro Manila (Gil Puyat and Guadalupe branches). The platform supports real-time bed inventory tracking, 5-step guided reservations, tamper-evident digital lease execution, pro-rata electricity and water sub-meter utility billing, PayMongo payment gateway integration, maintenance ticketing with turnaround-time tracking, and structured offboarding.

1.1 System Roles & Granular Permissions

Role	Portal Scope	Core Responsibilities & Capabilities
Guest	Public Pages	Browse room types, explore branch amenities, inspect transparent pricing, view bed availability, and submit pre-booking inquiries.
Applicant	Self-Service Onboarding	Register account, complete 5-step room/bed reservation, declare personal appliances, upload identification documents, and pay reservation fee.
Tenant	Tenant Portal	Review and digitally sign lease agreements, view itemized monthly rent & utility statements, pay online via PayMongo, submit maintenance tickets, and chat with management.
Admin	Assigned Branch Portal	Review and approve reservations, conduct physical check-in and key issuance, enter sub-meter readings, generate billing invoices, assign maintenance technicians, and post announcements.
Owner	Executive Platform	Supervise branch administrators, inspect cross-branch financial reports, manage platform pricing policies, review immutable audit trails, and oversee system governance.

2. Agile User Stories Catalog

The following catalog establishes the product requirements framed in Agile format. Each user story details user role, intended action, expected business value, acceptance criteria, story points, and mapped test cases.

EPIC 1: Public Discovery & Inquiry Management

US-01: Public Room & Bed Availability Browsing

Role: Guest

Priority: High

5 Pts

Trace: TC-01

"As a Guest, I want to browse available rooms filtered by branch, room type, and price range, so that I can find accommodations matching my preferences."

ACCEPTANCE CRITERIA

- Filtering by branch (Gil Puyat / Guadalupe) dynamically updates visible room inventory.
- Room cards accurately show real-time bed indicators (Available, Reserved, Occupied).
- Rooms under maintenance or marked inactive are automatically excluded from public results.

US-02: Pre-Booking Guest Inquiry Submission

Role: Guest

Priority: Medium

3 Pts

Trace: TC-02

"As a Guest, I want to submit an inquiry regarding room policies and scheduling, so that staff can clarify my questions prior to booking."

ACCEPTANCE CRITERIA

- Submitting full name, valid email, phone, and inquiry text triggers database persistence.
- Automated confirmation email dispatched immediately to the guest.
- Inquiry record renders instantly on the assigned Branch Admin Inquiry Dashboard.

EPIC 2: Room Reservation & Bed Selection Workflow

US-03: Bed Selection & Appliance Declaring

Role: Applicant

Priority: High

8 Pts

Trace: TC-03

"As an Applicant, I want to pick a specific bed and declare electrical appliances, so that my monthly billing and initial deposit are computed transparently."

ACCEPTANCE CRITERIA

- Applicant chooses between upper/lower bunk or single beds with real-time availability check.
- Selecting appliance checkboxes (fan, mini-fridge, laptop) adds standard fee tariffs to checkout preview.
- System strictly enforces a maximum limit of one active pending reservation per applicant.

US-04: 5-Step Guided Reservation Application

Role: Applicant

Priority: High

13 Pts

Trace: TC-04

"As an Applicant, I want to complete a guided 5-step onboarding wizard, so that I can schedule an on-site visit, submit verification documents, and pay the reservation fee."

ACCEPTANCE CRITERIA

- Step 1 (Summary), Step 2 (Visit/Rules), Step 3 (Application/ID Upload), Step 4 (Payment/Proof), Step 5 (Confirmation).
- Validates uploaded document mime-types and sizes (PNG, JPG, PDF up to 5MB).
- Generates a unique tracking Reservation Code upon final submission.

US-05: Atomic Bed Concurrency & Collision Lock

Role: System / Security

Priority: High

8 Pts

Trace: TC-05

"As a System Administrator, I want atomic booking transactions on bed records, so that simultaneous booking attempts on the same bed never result in double-booking."

ACCEPTANCE CRITERIA

- When two transactions target the same Bed ID concurrently, the first transaction acquires the bed lock.
- The secondary transaction rolls back cleanly, returning an HTTP 409 Conflict with a clear user prompt.

EPIC 3: Administrative Verification & Application Review

US-06: Admin Review & Reservation Approval

Role: Admin

Priority: High

5 Pts

Trace: TC-06

"As an Admin, I want to review uploaded applicant identification and payment verification, so that I can approve qualified occupants and initiate their contract."

ACCEPTANCE CRITERIA

- Admin verifies credentials on the admin portal and clicks 'Approve'.
- Status updates to 'approved', and the system initializes a pending digital lease agreement.
- Applicant receives an email containing contract review and digital signature instructions.

US-07: Reservation Rejection & Bed Release

Role: Admin

Priority: Medium

3 Pts

Trace: TC-07

"As an Admin, I want to reject invalid or incomplete applications with a mandatory reason, so that the reserved bed is immediately released back to the public pool."

ACCEPTANCE CRITERIA

- Admin submits rejection reason; reservation status changes to 'rejected'.
- Reserved bed transitions atomically from 'reserved' back to 'available'.
- Applicant receives an automated notification detailing the reason for rejection.

EPIC 4: Digital Lease Contracts & E-Signatures

US-08: Tenant Contract Review & Digital E-Signing

Role: Tenant

Priority: High

8 Pts

Trace: TC-08

"As a Tenant, I want to review my digital lease contract and sign electronically, so that my residency agreement is established legally without paper forms."

ACCEPTANCE CRITERIA

- Displays contract terms, room monthly rate, security deposit value, and house rules.
- Captures digital signature stroke data from touch/mouse canvas pad.
- Generates tamper-evident PDF document hash with signer IP address and timestamp.

US-09: Admin Final Countersign & Contract Activation

Role: Admin

Priority: High

5 Pts

Trace: TC-09

"As an Admin, I want to finalize and countersign the tenant's signed lease, so that residency activation can proceed."

ACCEPTANCE CRITERIA

- Admin verifies tenant e-signature and clicks 'Finalize Contract'.
- Contract status transitions to 'active' and official canonical PDF is permanently archived.

EPIC 5: Tenant Check-In & Residency Activation

US-10: Physical Check-In & Ledger Initialization

Role: Admin

Priority: High

8 Pts

Trace: TC-10

"As an Admin, I want to log physical check-in and key turnover, so that the tenant's residency is activated and recurring rent billing begins."

ACCEPTANCE CRITERIA

- Admin clicks 'Execute Check-In' and inputs actual move-in timestamp.
- User status transitions to 'checked-in' tenant and bed status transitions to 'occupied'.
- Initial billing ledger is generated and monthly rent cycle is scheduled.

EPIC 6: Sub-Meter Utility Billing & PayMongo Checkout

US-12: Sub-Meter Utility Reading & Pro-Rata Split Engine

Role: Admin / System

Priority: Critical

13 Pts

Trace: TC-12, TC-13

"As an Admin, I want to enter monthly electricity and water sub-meter readings for each room, so that charges are calculated and split pro-rata among roommates based on active occupancy days."

ACCEPTANCE CRITERIA

- Admin enters previous and present readings; consumption = Present - Previous.
- Calculates total room cost using official branch electricity (kWh) and water rates.
- Splits costs proportionally based on days each roommate was active during the cycle.
- Enforces precondition: only tenants with 'checked-in' status receive billing items.

US-13: Itemized Tenant Billing Statement

Role: Tenant

Priority: High

5 Pts

Trace: TC-13

"As a Tenant, I want to view an itemized monthly statement showing rent, electricity, water, appliance fees, and due date, so that I understand all charges."

ACCEPTANCE CRITERIA

- Billing portal displays clear breakdown with status badges (Pending, Paid, Overdue).
- Tenant can download official PDF Statement of Account with branch remittance details.

US-14: Online Payment via PayMongo & Automatic Reconciliation

Role: Tenant

Priority: High

8 Pts

Trace: TC-14

"As a Tenant, I want to pay my monthly rent and utilities via PayMongo (GCash, Maya, Cards), so that my payment settles immediately without manual verification delays."

ACCEPTANCE CRITERIA

- Checkout payload exact amount matches database bill balance to prevent price drift.
- Verified PayMongo webhook transitions bill status to 'Paid' instantly.
- Automated digital payment receipt is generated and tenant balance reflects zero.

EPIC 7: Maintenance Ticketing & Turnaround Time Tracking

US-15: Tenant Maintenance Request Submission

Role: Tenant

Priority: High

5 Pts

Trace: TC-15

"As a Tenant, I want to submit repair tickets with photos, categories, and urgency levels, so that facility issues are fixed promptly."

ACCEPTANCE CRITERIA

- Category options: Plumbing, Electrical, Hardware, Appliance, Cleaning.
- Supports image attachment upload and displays target turnaround time.
- Admin dashboard immediately surfaces ticket with urgency badge.

US-16: Admin Maintenance Assignment & Status Updates

Role: Admin

Priority: Medium

5 Pts

Trace: TC-16

"As an Admin, I want to assign tickets to maintenance personnel and track resolution, so that tenants stay informed of repair progress."

ACCEPTANCE CRITERIA

- Status advances: Pending -> In-Progress -> Completed (or Cancelled).
- Tenant receives automated notification upon completion with admin resolution remarks.

EPIC 8: Community Announcements & In-App Communications

US-17: Branch Announcement Engine with Mandatory Acknowledgment

Role: Admin

Priority: Medium

5 Pts

Trace: TC-18

"As an Admin, I want to post categorized announcements requiring mandatory tenant acknowledgment, so that critical dormitory notices are confirmed."

ACCEPTANCE CRITERIA

- Categories: Maintenance, Policy, Event, Alert, General. Target: All, Branch, Tenants.
- High-priority alerts trigger an acknowledgment modal before tenant can proceed.
- Admin view provides full audit list of tenants who acknowledged the post.

US-18: Real-Time In-App Support Chat

Role: Tenant / Admin

Priority: Medium

8 Pts

Trace: TC-17

"As a Tenant, I want to message branch management in real time, so that I can resolve urgent concerns quickly."

ACCEPTANCE CRITERIA

- Direct socket-powered chat channel between tenant and branch admin.
- Real-time message synchronization with read receipts and timestamps.

EPIC 9: Room Transfers & Move-Out Offboarding

US-19: Scheduled Room Transfer with Deposit Preservation

Role: Tenant / Admin

Priority: High

8 Pts

Trace: TC-19

"As a Tenant, I want to transfer to another available bed or room while preserving my security deposit, so that I can relocate smoothly without re-applying."

ACCEPTANCE CRITERIA

- Tenant submits target room request; Admin approves transfer schedule.
- Atomically frees previous bed and locks new bed slot on transfer cutover.
- Security deposit balance transfers cleanly to the new lease agreement.

US-20: Move-Out Clearance & Security Deposit Settlement

Role: Admin

Priority: High

8 Pts

Trace: TC-20

"As an Admin, I want to record move-out inspections and compute net deposit refunds, so that tenant offboarding is settled transparently."

ACCEPTANCE CRITERIA

- Admin enters authorized deductions (unpaid utility balances or room repairs).
- System calculates Net Refund = Security Deposit - Total Deductions.
- Contract marked 'terminated', bed returned to available pool, tenant archived.

EPIC 10: Multi-Branch Governance & Immutable Audit Logs

US-21: Multi-Branch Financial & Occupancy Analytics

Role: Owner

Priority: High

5 Pts

Trace: TC-21

"As an Owner, I want cross-branch financial metrics and occupancy comparisons, so that I can monitor dorm performance and revenue health."

ACCEPTANCE CRITERIA

- Side-by-side branch comparison cards (Gil Puyat vs Guadalupe).
- Date range filtering supports 7d, 30d, 90d, 365d, and All-Time ('all') scope.
- Enables clean CSV export for external auditing and accounting records.

US-22: Immutable Executive Audit Trail Logging

Role: Owner / Security

Priority: High

5 Pts

Trace: TC-22

"As an Owner, I want an immutable audit log of all system mutations and authentication events, so that security breaches and unauthorized changes can be traced."

ACCEPTANCE CRITERIA

- Captures actor ID, IP address, timestamp, event severity, and state mutations.
- Audit logs cannot be modified or deleted through any portal interface.

3. Master Quality Assurance Test Plan

The testing plan provides a structured verification procedure covering all 14 operational phases of the dormitory management lifecycle. Each scenario specifies preconditions, precise test steps, expected operational outcomes, and evaluation sign-off.

TC ID	Phase & Scenario	Test Execution Steps	Expected Operational Result	Pass
TC-01	Phase 1: Room Discovery & Filtering	1. Navigate to /check-availability. 2. Select Gil Puyat branch. 3. Filter by 2-Bed Shared room type. 4. Inspect room card results.	Only active rooms with available beds matching filter criteria display. Occupied beds indicated accurately. Inactive rooms hidden.	[]
TC-02	Phase 1: Guest Inquiry Submission	1. Click 'Inquire' on room card. 2. Fill name, email, phone, inquiry message. 3. Submit inquiry form.	Success notification appears. Automated confirmation dispatched to guest email. Admin dashboard receives new inquiry.	[]
TC-03	Phase 2: Bed Selection & Appliance Fee	1. Select Bed B in Room 102. 2. Check Mini-refrigerator and Fan appliances. 3. Verify monthly pricing summary.	Bed temporarily reserved during session. Total monthly rent recalculates dynamically to include appliance tariffs.	[]
TC-04	Phase 2: 5-Step Guided Reservation	1. Step 1: Confirm room details. 2. Step 2: Schedule visit & accept rules. 3. Step 3: Input personal info & upload ID. 4. Step 4: Submit payment proof. 5. Step 5: Receive code.	Reservation state saved as pending. Target bed marked reserved. Unique tracking code generated.	[]
TC-05	Phase 2: Bed Booking Concurrency Lock	1. Open two browser sessions with different applicants. 2. Select exact same Bed A in Room 204. 3. Submit reservation simultaneously.	First request succeeds. Second request rolls back cleanly with HTTP 409 Conflict and friendly re-selection prompt.	[]
TC-06	Phase 3: Admin Review & Approval	1. Log in as Branch Admin. 2. Open /admin/reservations. 3. Inspect uploaded documents. 4. Click Approve .	Status updates to approved. System initializes draft lease agreement. Notification email dispatched to applicant.	[]
TC-07	Phase 3: Admin Rejection & Bed Release	1. Open pending reservation. 2. Click Reject . 3. Input mandatory rejection reason. 4. Confirm rejection.	Status updates to rejected. Reserved bed returns atomically to public available pool.	[]
TC-08	Phase 4: Contract Review & E-Signing	1. Log in as approved applicant. 2. Navigate to /contracts. 3. Review terms & draw electronic signature. 4. Submit signature.	Signature coordinates, IP address, and timestamp hash saved. Contract status changes to signed. Canonical PDF ready.	[]
TC-09	Phase 4: Admin Final Countersign	1. Admin opens /admin/contracts. 2. Review applicant signature. 3. Click Finalize Contract .	Contract status transitions to active. Tamper-evident PDF permanently archived in system storage.	[]
TC-10	Phase 5: Check-In & Residency Activation	1. Open Tenant Details in admin portal. 2. Hand over keys. 3. Click Execute Check-In with actual timestamp.	User role changes to checked-in tenant. Bed state changes to occupied. Monthly billing cycle starts.	[]
TC-11	Phase 6: Profile Security & Locked Fields	1. Tenant opens /profile. 2. Update phone and emergency contact. 3. Attempt to alter assigned room via API payload.	Contact details update smoothly. Room number alteration blocked with HTTP 403 Forbidden by profile security lock.	[]

TC ID	Phase & Scenario	Test Execution Steps	Expected Operational Result	Pass
TC-12	Phase 7: Sub-Meter Utility Billing Split	1. Enter electricity meter: 100 kWh consumed. 2. Tariff = PHP 14.00/kWh (Total = PHP 1,400). 3. Room has 2 full-month checked-in tenants.	Engine computes room total PHP 1,400 and splits exactly PHP 700.00 to each tenant's monthly billing statement.	[]
TC-13	Phase 7: Pro-Rata Split on Mid-Cycle Move-In	1. Room consumption = 150 kWh. 2. Tenant A active 30 days; Tenant B moved in mid-cycle (active 15 days). 3. Generate utility split.	Engine factors active days proportionally. Tenant A billed 2/3 (PHP 1,400) and Tenant B billed 1/3 (PHP 700) accurately.	[]
TC-14	Phase 8: PayMongo Online Payment Checkout	1. Tenant clicks Pay with PayMongo on bill. 2. Settle via GCash test gateway. 3. PayMongo webhook dispatches.	Bill status marks Paid. Official digital receipt generated. Tenant ledger balance drops to PHP 0.00.	[]
TC-15	Phase 9: Maintenance Request Submission	1. Tenant opens /maintenance. 2. Select category 'Plumbing', urgency 'High'. 3. Upload photo and submit.	Ticket created with unique ID. Target turnaround time displayed. Real-time alert dispatched to branch admin.	[]
TC-16	Phase 9: Maintenance Assignment & Resolution	1. Admin assigns technician to ticket. 2. Advance status to In-Progress. 3. Complete repair and log resolution remarks.	Ticket status updates to Completed. Tenant receives resolution notification with remarks.	[]
TC-17	Phase 10: Real-Time In-App Support Chat	1. Tenant sends message in chat portal. 2. Admin views message in real time and replies.	Messages transmit instantaneously via Socket.io without page refresh. Read status markers update upon opening.	[]
TC-18	Phase 11: Emergency Announcement Modal	1. Admin posts 'Alert' announcement. 2. Target set to 'All Tenants'. 3. Tenant accesses dashboard.	Mandatory modal overlays screen. Tenant clicks 'Acknowledge'. Admin view logs acknowledgment timestamp.	[]
TC-19	Phase 12: Room Transfer & Deposit Preservation	1. Tenant applies for room transfer. 2. Admin approves new room assignment. 3. Execute transfer cutover.	Previous bed released, new bed marked occupied. Existing security deposit transfers intact without penalty.	[]
TC-20	Phase 13: Move-Out Clearance & Net Refund	1. Admin conducts room inspection. 2. Enters deduction of PHP 500 for utility balance. 3. Approves final settlement.	Net refund calculated accurately. Contract marked terminated. Bed released back to public available pool.	[]
TC-21	Phase 14: Cross-Branch Owner Analytics	1. Log in as Dorm Owner. 2. Open Analytics dashboard. 3. Select 'All-Time' scope and export CSV.	Consolidated revenue and occupancy percentages render accurately. CSV file downloads with complete data fields.	[]
TC-22	Phase 14: Executive Audit Trail Verification	1. Admin modifies room pricing. 2. Owner opens /admin/audit-logs.	Audit record appears displaying Admin ID, timestamp, IP address, target resource, and before/after mutation diff.	[]

4. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix links each Agile User Story directly to its operational Test Case ID, backend Jest integration test file, and test automation coverage.

Story ID	Feature / Capability	Test Case	Automated Jest Test Suite / Controller	Coverage
US-01	Room & Availability Browsing	TC-01	server/controllers/roomsController.test.js	100%
US-02	Pre-Booking Guest Inquiry	TC-02	server/controllers/inquiriesController.create.test.js	100%
US-03	Bed Selection & Appliance Declaration	TC-03	server/controllers/reservations/_helpers.appliance.test.js	100%
US-04	5-Step Guided Reservation Application	TC-04	server/controllers/reservationsController.test.js	100%
US-05	Atomic Bed Booking Concurrency Lock	TC-05	server/controllers/reservationHelpers.test.js	100%
US-06	Admin Reservation Review & Approval	TC-06	server/controllers/reservationsController.access.test.js	100%
US-07	Admin Rejection & Bed Release	TC-07	server/controllers/reservationsController.test.js	100%
US-08	Tenant Contract Review & E-Signing	TC-08	server/controllers/contractSigningWiring.test.js	100%
US-09	Admin Final Contract Countersign	TC-09	server/controllers/contractFinalDocumentReplacementWiring.test.js	100%
US-10	Tenant Check-In & Residency Activation	TC-10	server/controllers/reservationsController.test.js	100%
US-11	Room Grid & Inventory Management	TC-11	server/controllers/roomsController.test.js	100%
US-12	Sub-Meter Utility Split Calculation	TC-12	server/controllers/monthlyUtilityWorkflow.integration.test.js	100%
US-13	Itemized Tenant Billing Statement	TC-13	server/controllers/billingController.test.js	100%
US-14	PayMongo Online Payment Reconciliation	TC-14	server/controllers/paymentController.test.js	100%
US-15	Maintenance Ticket Submission	TC-15	server/controllers/maintenanceController.test.js	100%

Story ID	Feature / Capability	Test Case	Automated Jest Test Suite / Controller	Coverage
US-16	Maintenance Resolution & Assignment	TC-16	server/controllers/maintenanceController.test.js	100%
US-17	Announcement Engine & Acknowledgment	TC-18	server/controllers/announcementsController.test.js	100%
US-18	In-App Support Chat & Real-Time Comms	TC-17	server/controllers/chatController.context.test.js	100%
US-19	Room Transfer & Deposit Preservation	TC-19	server/controllers/roomTransferController.test.js	100%
US-20	Move-Out Clearance & Net Refund	TC-20	server/controllers/contractLifecycle.test.js	100%
US-21	Owner Multi-Branch Analytics	TC-21	server/controllers/analyticsController.test.js	100%
US-22	Executive Audit Trail Verification	TC-22	server/controllers/auditController.test.js	100%

5. Quality Assurance Defense & Verification Sign-Off

The undersigned attest that all functional requirements, security guards, concurrency locks, and operational test procedures documented herein have been inspected, tested, and validated against the Lilycrest Dormitory Management System codebase.

Evaluation Role	Designated Name	Verdict	Signature & Date
Lead Software Engineer	Capstone Team Lead	[] Approved & Passed	_____
Quality Assurance Specialist	QA / Test Engineer	[] Approved & Passed	_____
Lead System Architect	Backend & Database Specialist	[] Approved & Passed	_____
Faculty Adviser / Panelist	Capstone Project Adviser	[] Approved & Passed	_____